

Geometry-based Methods in Vision

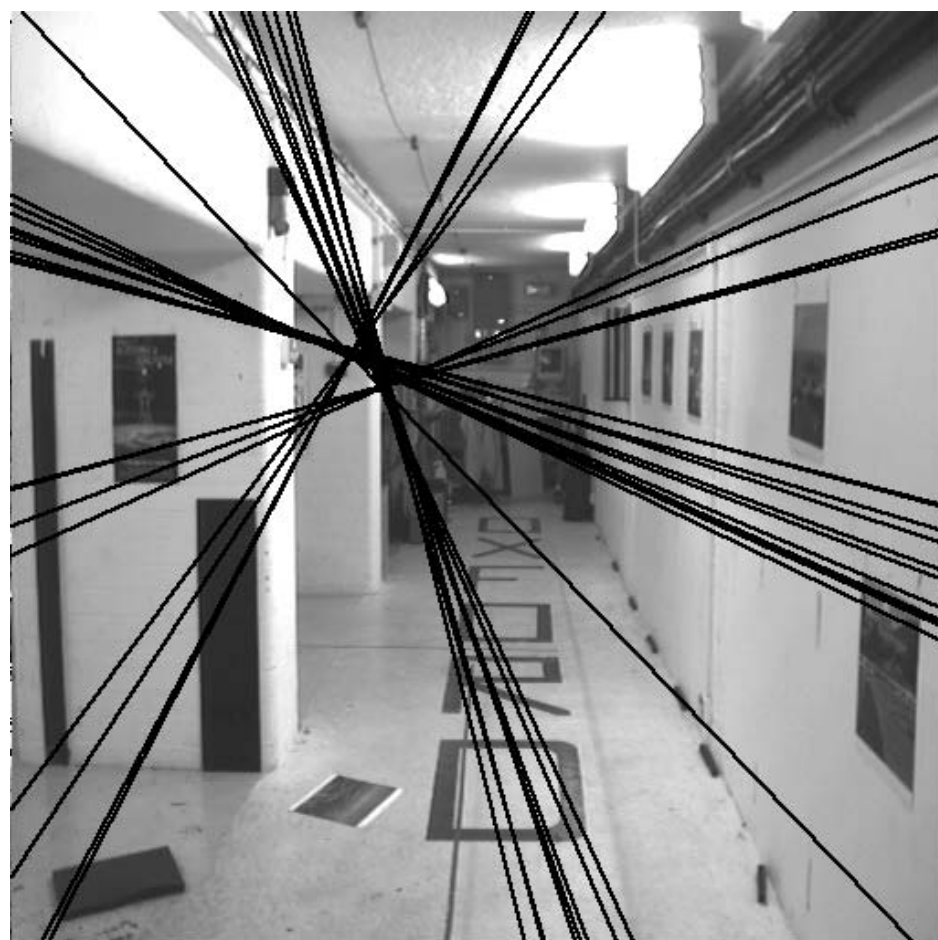
CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

Administrivia

- Problem Set 2
 - Due **today**
- Assignment 2
 - Due on Tue next week
- Projects
 - Discussion in next lecture on expectations
 - Proposals due on Thu next week

Lecture 11: Two-view Reconstruction II



A reminder to ask questions!



Two-view Reconstruction



Input: Two images of a scene



Output: (Sparse) 3D
Reconstruction + Camera Poses

Two-view Reconstruction: A simplified approach



Step 1: Annotate/extract correspondences



Step 2: Compute F (uncalibrated case) or E (calibrated case)



Step 3: Recover projection matrices P, P'



Step 4: Compute 3D via triangulation

previous lecture

From Cameras and Correspondences to 3D



$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \quad (\mathbf{P}, \mathbf{P}')$$

$$\{\mathbf{X}_i\}$$

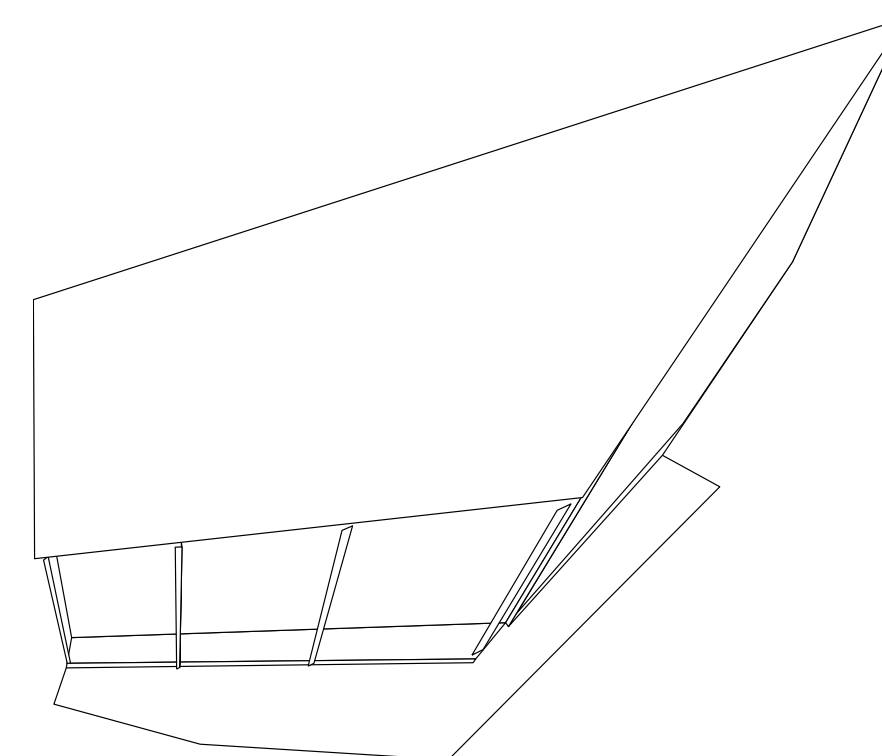
Camera Ambiguities imply 3D Ambiguities

$$\begin{array}{ccc} \{(\mathbf{x}_i, \mathbf{x}'_i)\} & (\mathbf{P}, \mathbf{P}') & \longrightarrow \{\mathbf{X}_i\} \\ \{(\mathbf{x}_i, \mathbf{x}'_i)\} & (\mathbf{P}H, \mathbf{P}'H) & \longrightarrow \{H^{-1}\mathbf{X}_i\} \end{array}$$

Different estimated cameras will yield corresponding different 3D structure

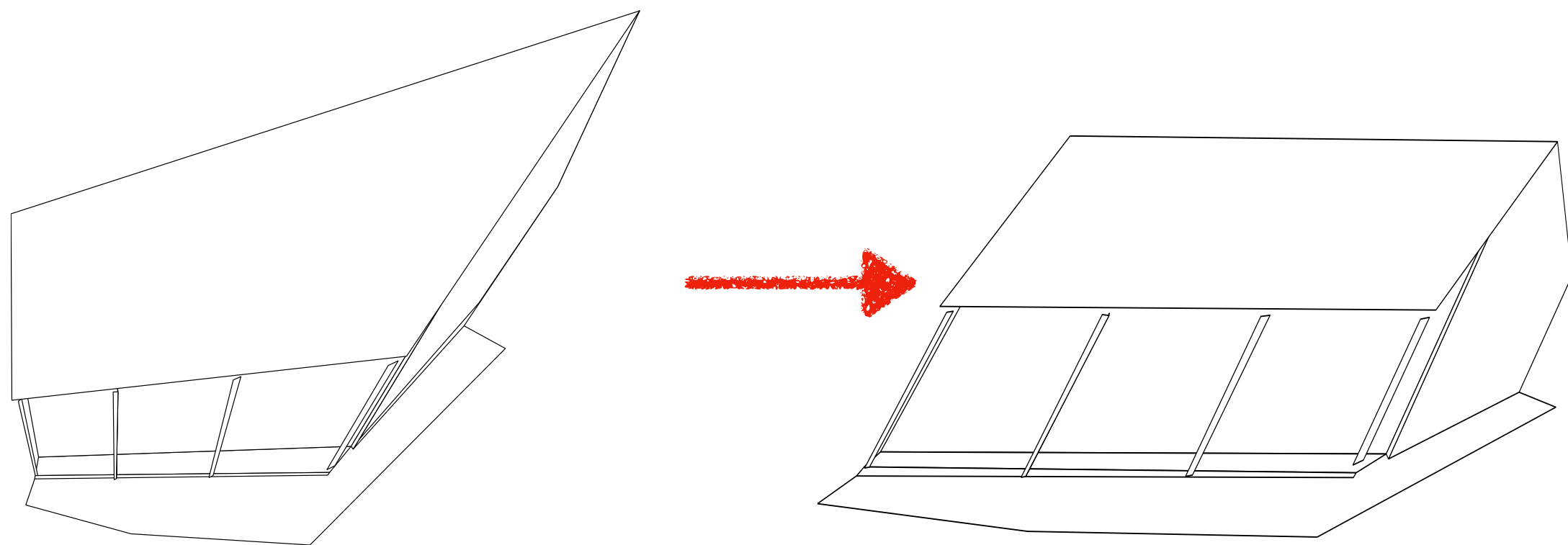
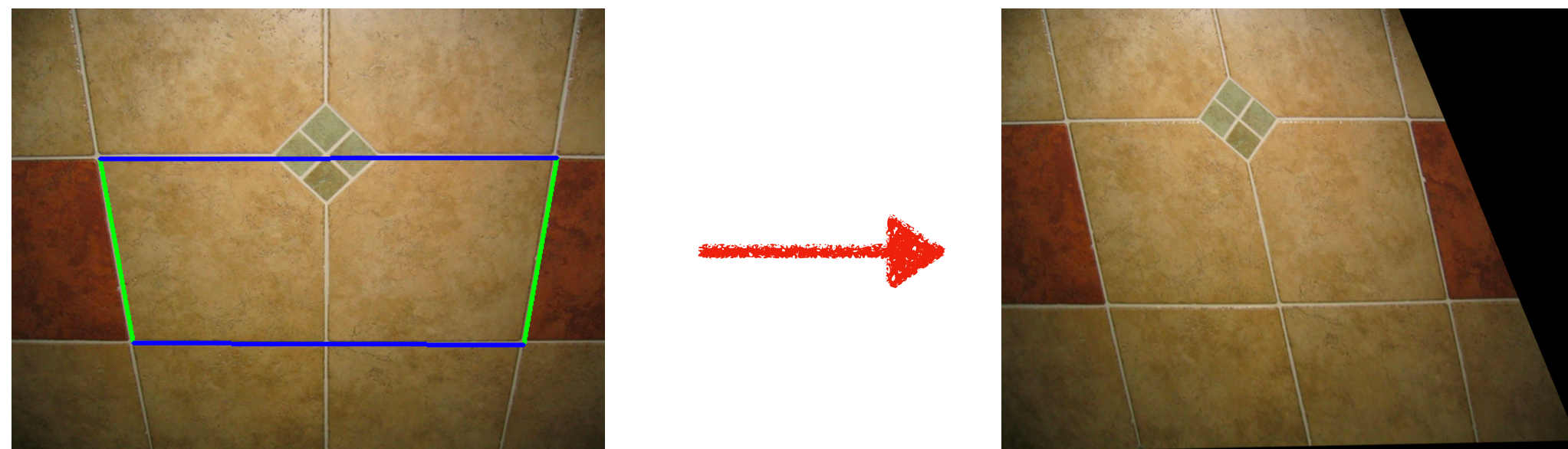
Calibrated case: can only recover 3D up to a similarity transform as $(\mathbf{P}, \mathbf{P}')$ are ambiguous up to a similarity

Uncalibrated case: can only recover 3D up to a homography as $(\mathbf{P}, \mathbf{P}')$ are ambiguous up to a general H



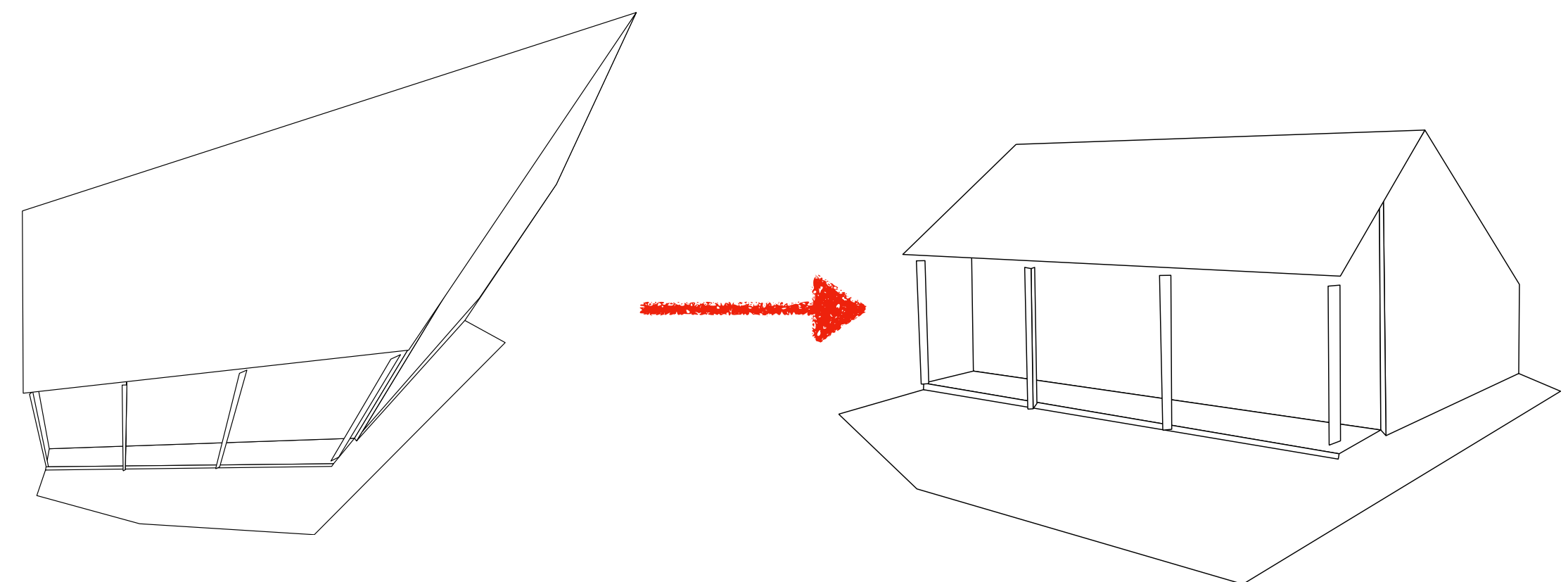
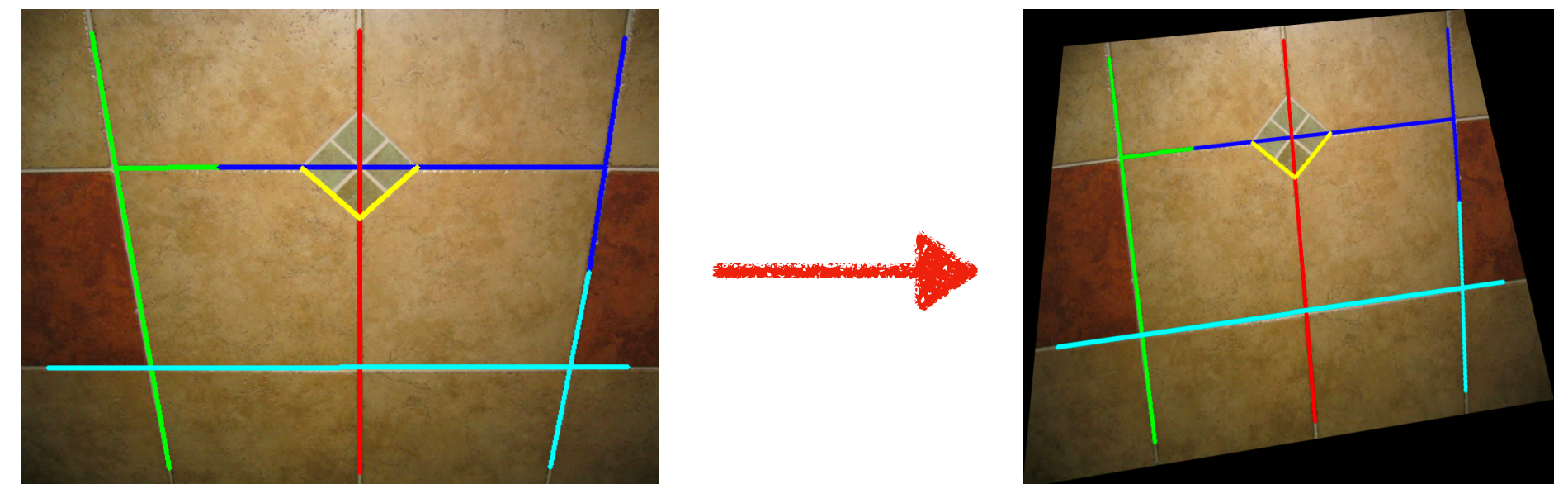
Annotation-based 3D Rectification

Affine rectification via annotating
parallel lines/directions



Compute H s.t. the line (plane) at infinity
is transformed to ideal position

Metric rectification via annotating
perpendicularity



Compute H that respects orthogonality of
lines (or 3D directions)

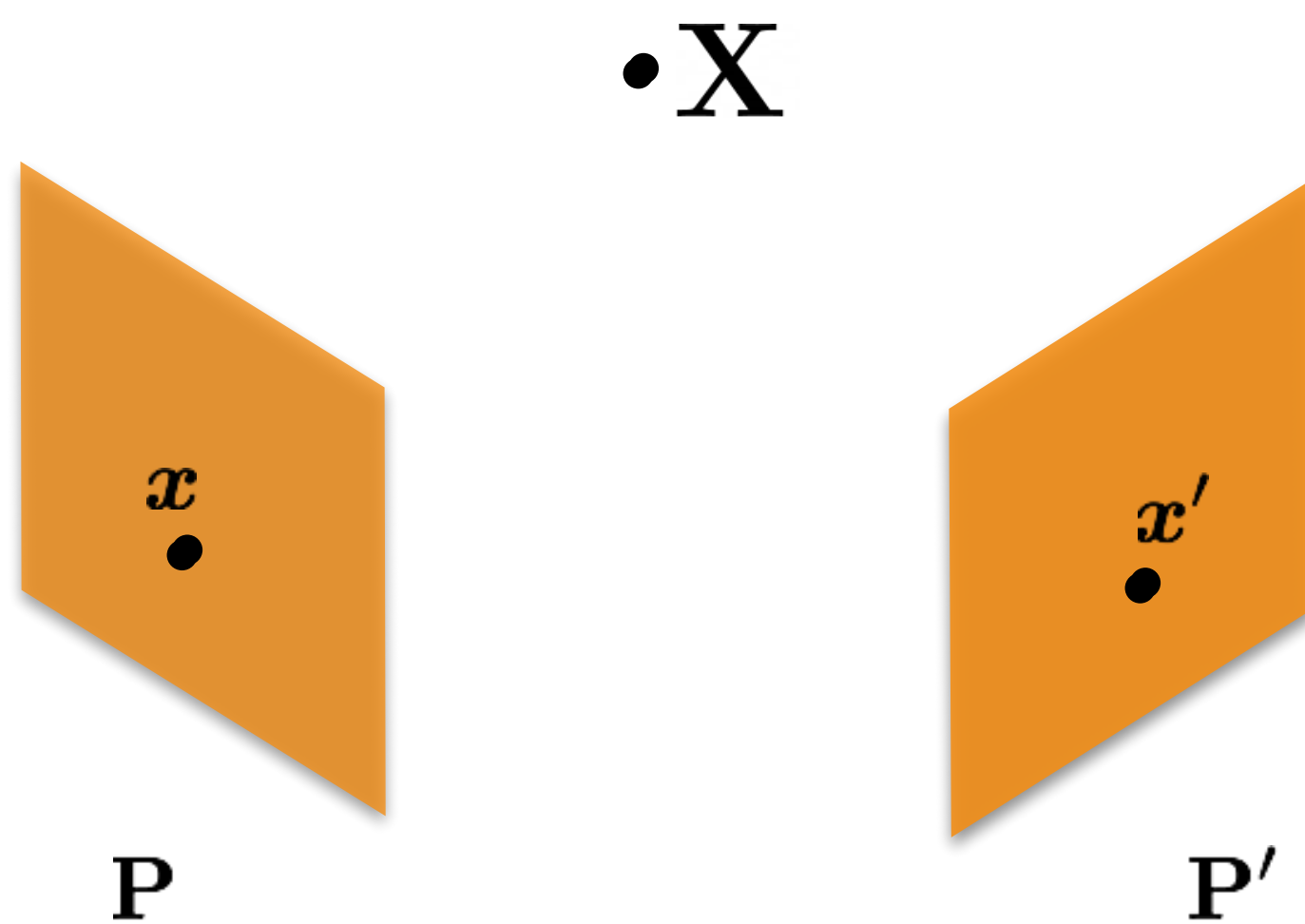
From Cameras and Correspondences to 3D



$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \quad (\mathbf{P}, \mathbf{P}')$$

$$\{\mathbf{X}_i\}$$

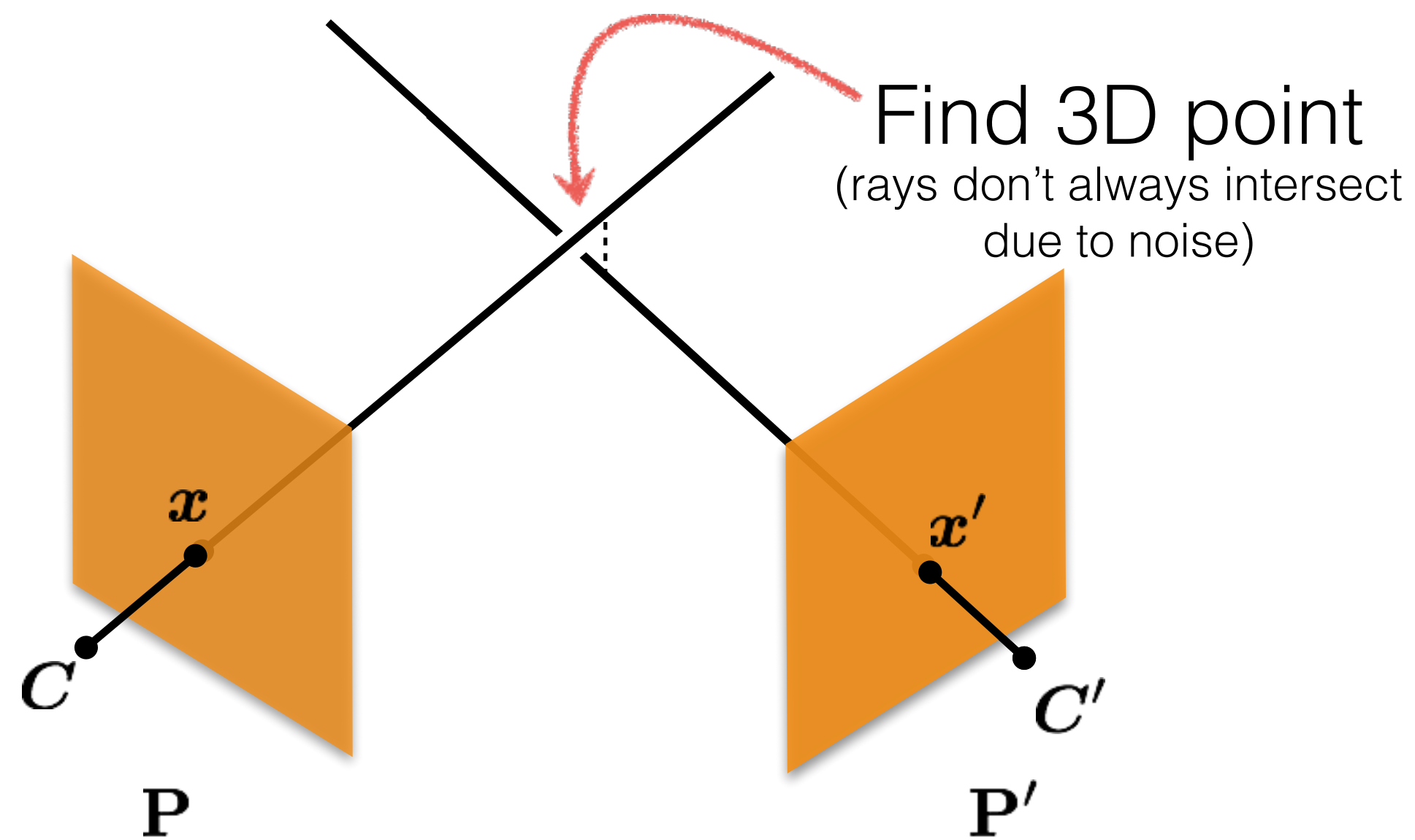
Triangulation



X s.t.

$$\mathbf{x} = \mathbf{P}\mathbf{X}, \quad \mathbf{x}' = \mathbf{P}'\mathbf{X}$$

Triangulation



$\mathbf{X}$ s.t.

$$\mathbf{x} = \mathbf{P}\mathbf{X}, \quad \mathbf{x}' = \mathbf{P}'\mathbf{X}$$

$$\mathbf{x} \times \mathbf{P}\mathbf{X} = \mathbf{0}$$

$$[\mathbf{x}]_{\times} \mathbf{P} \mathbf{X} = \mathbf{0}$$

$$\mathbf{A} \mathbf{X} = \mathbf{0}$$

$$\mathbf{x}' \times \mathbf{P}'\mathbf{X} = \mathbf{0}$$

$$[\mathbf{x}']_{\times} \mathbf{P}' \mathbf{X} = \mathbf{0}$$

$$\mathbf{A}' \mathbf{X} = \mathbf{0}$$

$$\begin{bmatrix} [\mathbf{x}]_{\times} \mathbf{P} \\ [\mathbf{x}']_{\times} \mathbf{P}' \end{bmatrix} \mathbf{X} = \mathbf{0}$$

6 equations, but only 4 are linearly independent (2 from each projection)

Disclaimer: efficient solution to implement, but does not minimize geometric reprojection loss

Two-view Reconstruction: A simplified approach



Step 1: Annotate/extract correspondences



Step 2: Compute F (uncalibrated case) or E (calibrated case)

Computing F : 8-pt algorithm,
7-pt algorithm

Computing E : 5-pt algorithm



Step 3: Recover projection matrices P, P'



Step 4: Compute 3D via triangulation

Estimating the Fundamental Matrix



$$\{(\mathbf{x}_i, \mathbf{x}'_i)\}$$

$$F$$

Q: What constraint can allow estimating F ?

$$\mathbf{x}'^\top F \mathbf{x} = 0$$

Estimating the Fundamental Matrix

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \longrightarrow F$$

$$\mathbf{x}'^\top F \mathbf{x} = 0$$

$$\mathbf{x} = (u, v, w) \quad \mathbf{x}' = (u', v', w')$$

$$\begin{bmatrix} u' & v' & w' \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = 0$$

$$f_{11}u'u + f_{12}u'v + f_{13}u'w + f_{21}v'u + f_{22}v'v + f_{23}v'w + f_{31}w'u + f_{32}w'v + f_{33}w'w = 0$$

Estimating the Fundamental Matrix

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \longrightarrow F$$

$$\mathbf{x}'^\top F \mathbf{x} = 0$$

$$\mathbf{A}_i \mathbf{f} = 0$$

$$\mathbf{A}_i = [u'_i u_i \quad u'_i v_i \quad u'_i w_i \quad v'_i u_i \quad v'_i v_i \quad v'_i w_i \quad w'_i u_i \quad w'_i v_i \quad w'_i w_i]$$

$$\mathbf{f} = [f_{11} \quad f_{12} \quad f_{13} \quad f_{21} \quad f_{22} \quad f_{23} \quad f_{31} \quad f_{32} \quad f_{33}]^\top$$

Q: How many constraints needed to estimate a general 3 X 3 homogenous matrix?

The 8-point Algorithm

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \longrightarrow F$$

Using 8 annotated/automatic correspondences, find:

$$\mathbf{f} \text{ s.t. } \mathbf{A}\mathbf{f} = \mathbf{0}; \mathbf{A} \in \mathbb{R}^{8 \times 9}$$

$$\mathbf{A}_i = [u'_i u_i \quad u'_i v_i \quad u'_i w_i \quad v'_i u_i \quad v'_i v_i \quad v'_i w_i \quad w'_i u_i \quad w'_i v_i \quad w'_i w_i]$$

$$\mathbf{f} = [f_{11} \quad f_{12} \quad f_{13} \quad f_{21} \quad f_{22} \quad f_{23} \quad f_{31} \quad f_{32} \quad f_{33}]^T$$

How to compute $\mathbf{f}$? Last singular vector of $\mathbf{A}$!

Does this yield a valid fundamental matrix F ?

The 8-point Algorithm: Enforcing Singularity

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \longrightarrow F$$

$$\mathbf{f} \text{ s.t. } \mathbf{A}\mathbf{f} = 0; \mathbf{A} \in \mathbb{R}^{8 \times 9}$$

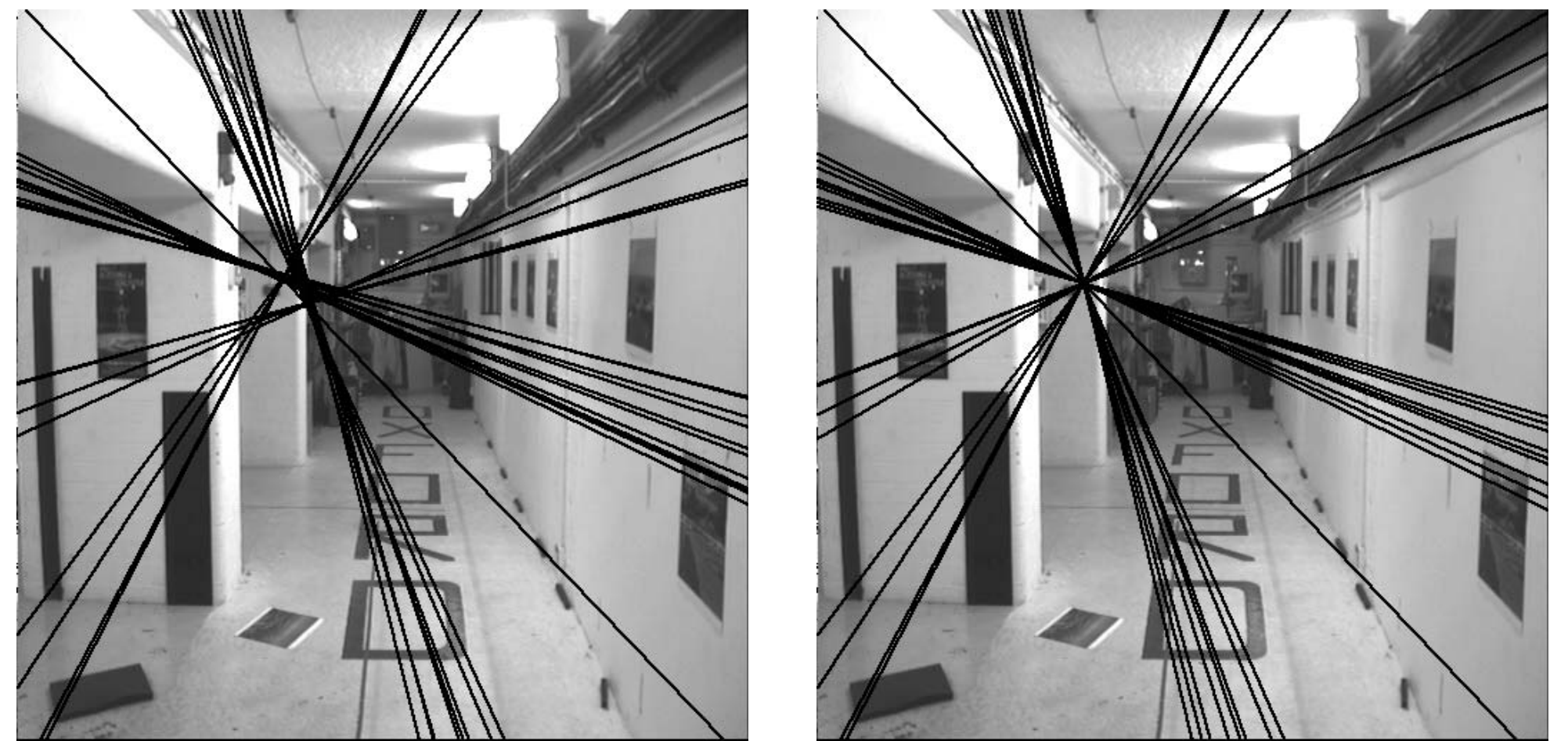
Enforcing $\det(F)=0$:

input:

$$F = \mathbf{U} \text{diag}(d_1, d_2, d_3) \mathbf{V}^\top$$

output:

$$F = \mathbf{U} \text{diag}(d_1, d_2, 0) \mathbf{V}^\top$$



Left: epipolar lines w/o singularity constraint.

Right: After enforcing $\det(F)=0$

Importance of Normalization

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \longrightarrow F$$

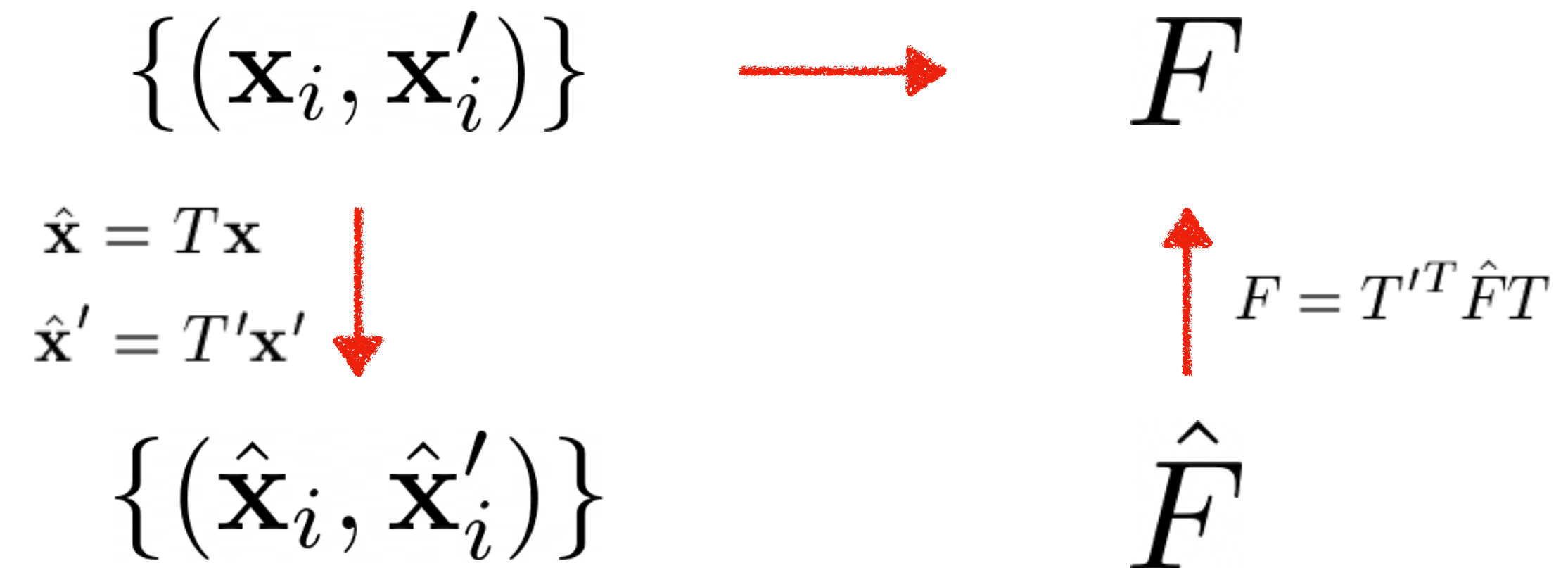
$$\mathbf{A}_i = \begin{bmatrix} u'_i u_i & u'_i v_i & u'_i w_i & v'_i u_i & v'_i v_i & v'_i w_i & w'_i u_i & w'_i v_i & w'_i w_i \end{bmatrix}$$

$= 100 \times 100$
 $= 1 \times 1$

$$\sim 10^{-4} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \sim 10^0$$

rank 2 approximation of F can have larger impact on some variables

Normalized 8-point Algorithm



A possible ‘normalization’ scheme is to zero-center and enforce unit variance

$$T = \begin{bmatrix} s & 0 & -sx_0 \\ 0 & s & -sy_0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(x_0, y_0) = \frac{1}{n} \sum_{i=1}^n (x_i, y_i) \quad s = \sqrt{2}/d_{avg}, \quad d_{avg} = \frac{1}{n} \sum_{i=1}^n \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2}$$

Normalized 8-point Algorithm: Summary

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \longrightarrow F$$

Step 1: Normalize data via a similarity transform

$$\hat{\mathbf{x}} = T\mathbf{x} \quad \hat{\mathbf{x}}' = T'\mathbf{x}'$$

Step 2: SVD using constraint matrix from 8 correspondences

$$\mathbf{f} \text{ s.t. } \mathbf{A}\mathbf{f} = 0; \mathbf{A} \in \mathbb{R}^{8 \times 9}$$

Step 3: Project F to rank 2 matrix

$$\mathbf{U} \operatorname{diag}(d_1, d_2, d_3) \mathbf{V}^\top \longrightarrow \mathbf{U} \operatorname{diag}(d_1, d_2, 0) \mathbf{V}^\top$$

Step 4: Transform solution to pixel space

$$F = T'^T \hat{F} T$$

Q: How would you modify this to compute the essential matrix E?

8-point Algorithm for computing E

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \xrightarrow{(K, K')} E$$

Step 1: Compute normalized coordinates

$$\hat{\mathbf{x}} = \mathbf{K}^{-1} \mathbf{x}, \quad \hat{\mathbf{x}}' = \mathbf{K}'^{-1} \mathbf{x}'$$

Step 2: SVD using constraint matrix from 8 correspondences

$$\mathbf{e} \text{ s.t. } \mathbf{A}\mathbf{e} = 0; \mathbf{A} \in \mathbb{R}^{8 \times 9}$$

Step 3: Project E to rank 2 matrix

$$\mathbf{U} \operatorname{diag}(d_1, d_2, d_3) \mathbf{V}^\top \xrightarrow{\quad} \mathbf{U} \operatorname{diag}(\bar{d}, \bar{d}, 0) \mathbf{V}^\top$$
$$\bar{d} = d_1/2 + d_2/2$$

Why do the singular values of E have this form?

RANSAC: Robust Inference under Outliers

- Do k times:
 - Draw set with minimum number of correspondences
 - Fit epipolar geometry (F or E) to the set
 - Count the number d of correspondences that are closer than t to the fitted epipolar lines
 - If $d > d_{\min}$, recompute fit error using all the correspondences
- Return best fit found

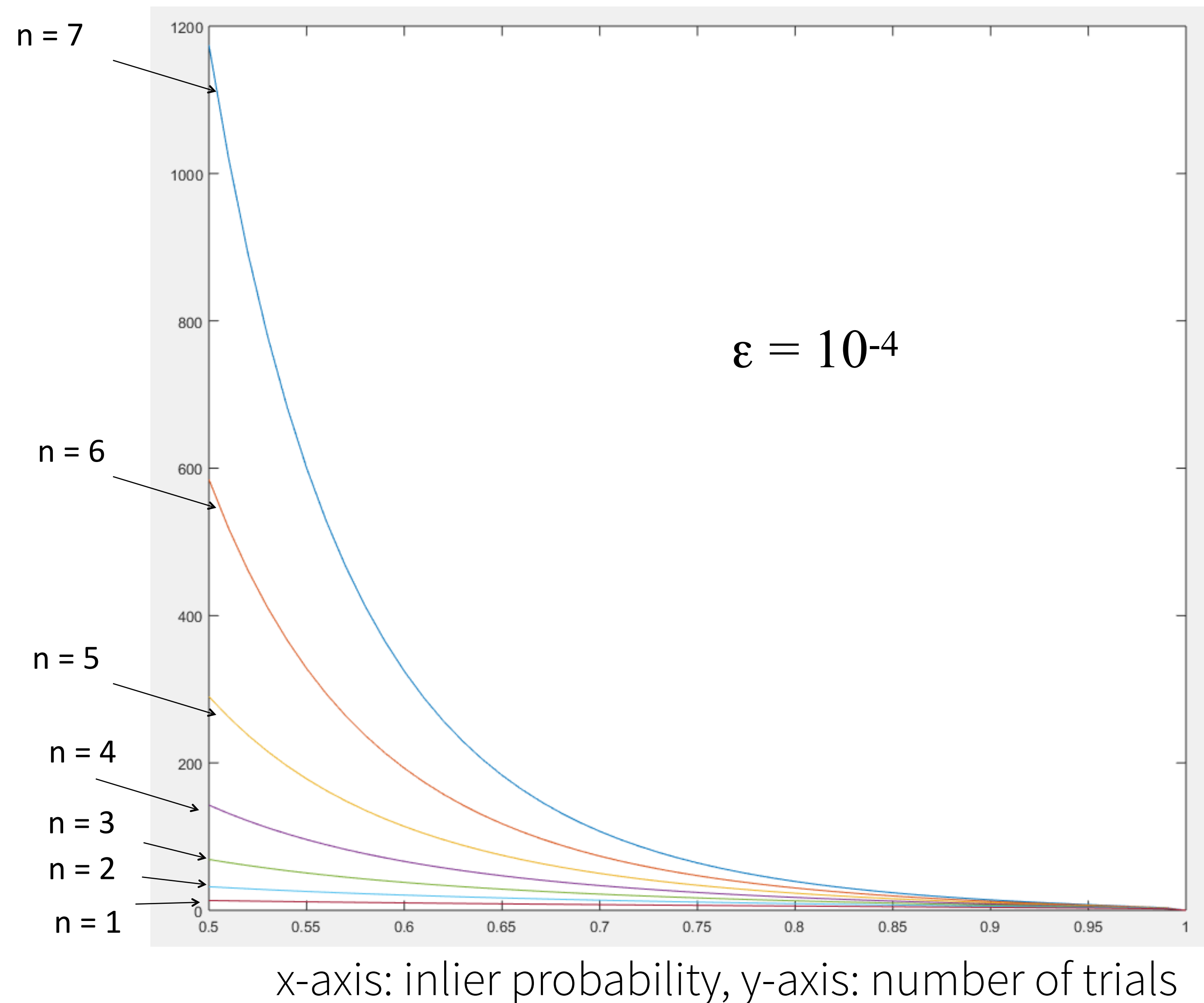
RANSAC: Effect of n in an n -point algorithm

- ε = max. probability of error
- t = minimum number of trials
- r = fraction of inliers

$$t = \frac{\log \varepsilon}{\log(1 - r^n)}$$

Require exponentially more trials to guarantee one run without any outliers!

Q: what is the minimal number of correspondences for computing F or E?



Computing F: 7-pt Algorithm

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \longrightarrow F$$

Using 7 annotated/automatic correspondences, find:

$$\mathbf{f} \text{ s.t. } \mathbf{A}\mathbf{f} = \mathbf{0}; \mathbf{A} \in \mathbb{R}^{7 \times 9}$$

Two linearly independent solutions:

$\mathbf{f}_1, \mathbf{f}_2$ — last two-singular vectors of $\mathbf{A}$

General solution for F:

$$\lambda F_1 + (1 - \lambda)F_2$$

Is there an additional constraint?

$$\lambda \text{ s.t. } \det(\lambda F_1 + (1 - \lambda)F_2) = 0$$

3-degree polynomial — at least one real solution!

(use `numpy.roots`)

(test all for inliers if multiple real solutions)

Computing E: 5-pt Algorithm

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \xrightarrow{(K, K')} E$$

Using 5 annotated/automatic correspondences, find:

$$\mathbf{e} \text{ s.t. } \mathbf{A}\mathbf{e} = 0; \mathbf{A} \in \mathbb{R}^{5 \times 9}$$

$$xE_1 + yE_2 + zE_3 + \underline{wE_4} \quad \text{set to 1}$$

Additional constraints on E?

$$\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{R} \quad E = \mathbf{U} \operatorname{diag}(\sigma, \sigma, 0) \mathbf{V}^{\top}$$

First 2 singular values should be equal, other
should be 0

Computing E: 5-pt Algorithm

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \xrightarrow{(K, K')} E$$

Using 5 annotated/automatic correspondences, find:

$$\mathbf{e} \text{ s.t. } \mathbf{A}\mathbf{e} = 0; \mathbf{A} \in \mathbb{R}^{5 \times 9}$$

$$xE_1 + yE_2 + zE_3 + \underline{wE_4} \quad \text{set to 1}$$

Equivalent conditions:

$$\det(E) = 0 \quad (EE^T - \frac{1}{2}\text{trace}(EE^T))E = 0$$

10 degree-3 polynomial equations in variables x,y,z
(one for each element of the above 3 X 3 matrix +
singularity constraint)

Computing E: 5-pt Algorithm

$$\begin{array}{c}
 \mathbf{C} \\
 \text{10 X 20}
 \end{array}
 \begin{array}{c}
 \left[\begin{array}{c} x^3 \\ y^3 \\ z^3 \\ x^2y \\ \vdots \\ z \\ 1 \end{array} \right] \\
 \text{20 X 1}
 \end{array}
 = \mathbf{0}
 \begin{array}{c}
 \text{10 X 1}
 \end{array}
 \xrightarrow{\hspace{1cm}}
 \begin{array}{c}
 \mathbf{C}(\mathbf{z}) \\
 \text{10 X 10}
 \end{array}
 \begin{array}{c}
 \left[\begin{array}{c} x^3 \\ y^3 \\ x^2y \\ \vdots \\ y \\ 1 \end{array} \right] \\
 \text{10 X 1}
 \end{array}
 = \mathbf{0}
 \begin{array}{c}
 \text{10 X 1}
 \end{array}$$

$$\det(\mathbf{C}(\mathbf{z})) = 0$$

(a 10-degree polynomial in z — can be efficiently solved via off-the-shelf solvers e.g. `numpy.roots`)

For each real-valued solution of z , compute (x,y) via the null-space of $\mathbf{C}(\mathbf{z})$

Suggested reading: Five-Point Motion Estimation Made Easy. *Li & Hartley*

Looking back: Computing F/E

$$\{(\mathbf{x}_i, \mathbf{x}'_i)\} \xrightarrow{\quad} F \qquad \{(\mathbf{x}_i, \mathbf{x}'_i)\} \xrightarrow{(K, K')} E$$

8-pt algorithm: can be used to uniquely compute either F or E

do we always get a unique solution given enough points? If no noise/outliers, would get correct solution directly

But in practice, need to project to space of valid E/F matrices

Towards more efficient solutions for RANSAC:

7-pt algorithm: determines F via solving a cubic equation (but multiple solutions may exist)

5-pt algorithm: determines E via solving a degree-10 equation (but multiple solutions may exist)

Degenerate Cases

Camera rotation



$$\mathbf{x}' = K R K^{-1} \mathbf{x}$$

$$\mathbf{x}^T H^T F \mathbf{x} = 0$$

For any skew-symmetric matrix S , the following is a solution

$$F = H^{-T} S$$

Thus, cannot determine F no matter how many annotations are used

All annotated points on a common plane



$$\mathbf{x}' = H \mathbf{x}$$

Two-view Reconstruction



Step 1: Annotate/extract correspondences



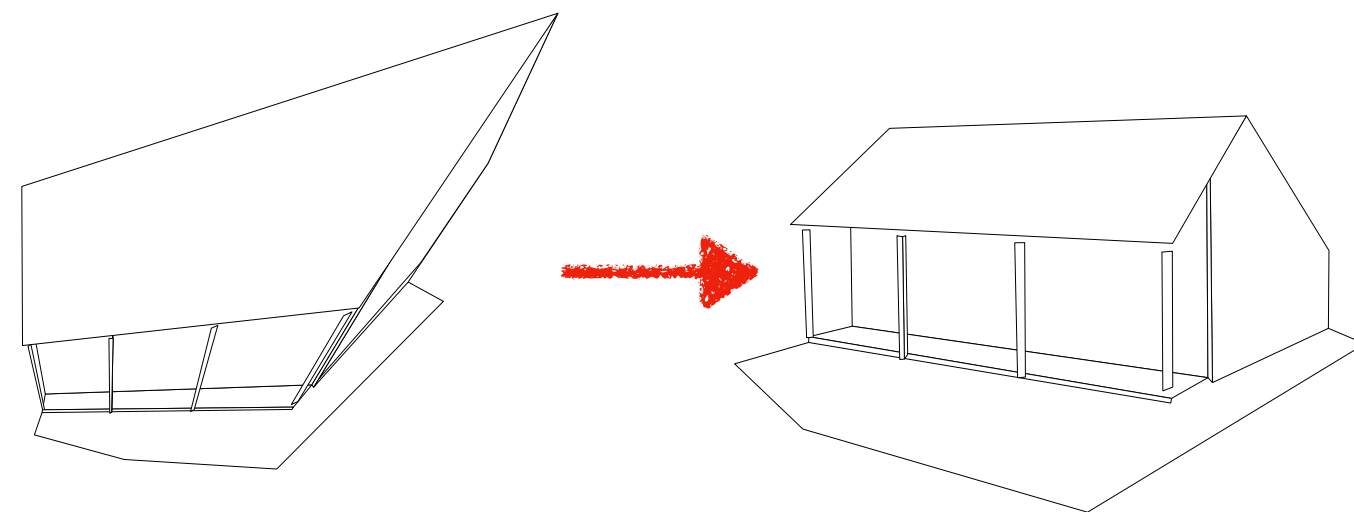
Step 2: Compute F (uncalibrated case) or E (calibrated case)



Step 3: Recover projection matrices P, P'



Step 4: Compute 3D via triangulation



+ rectification of 3D reconstruction

Next lecture: Iterative improvement

+ discussion on projects

Questions?